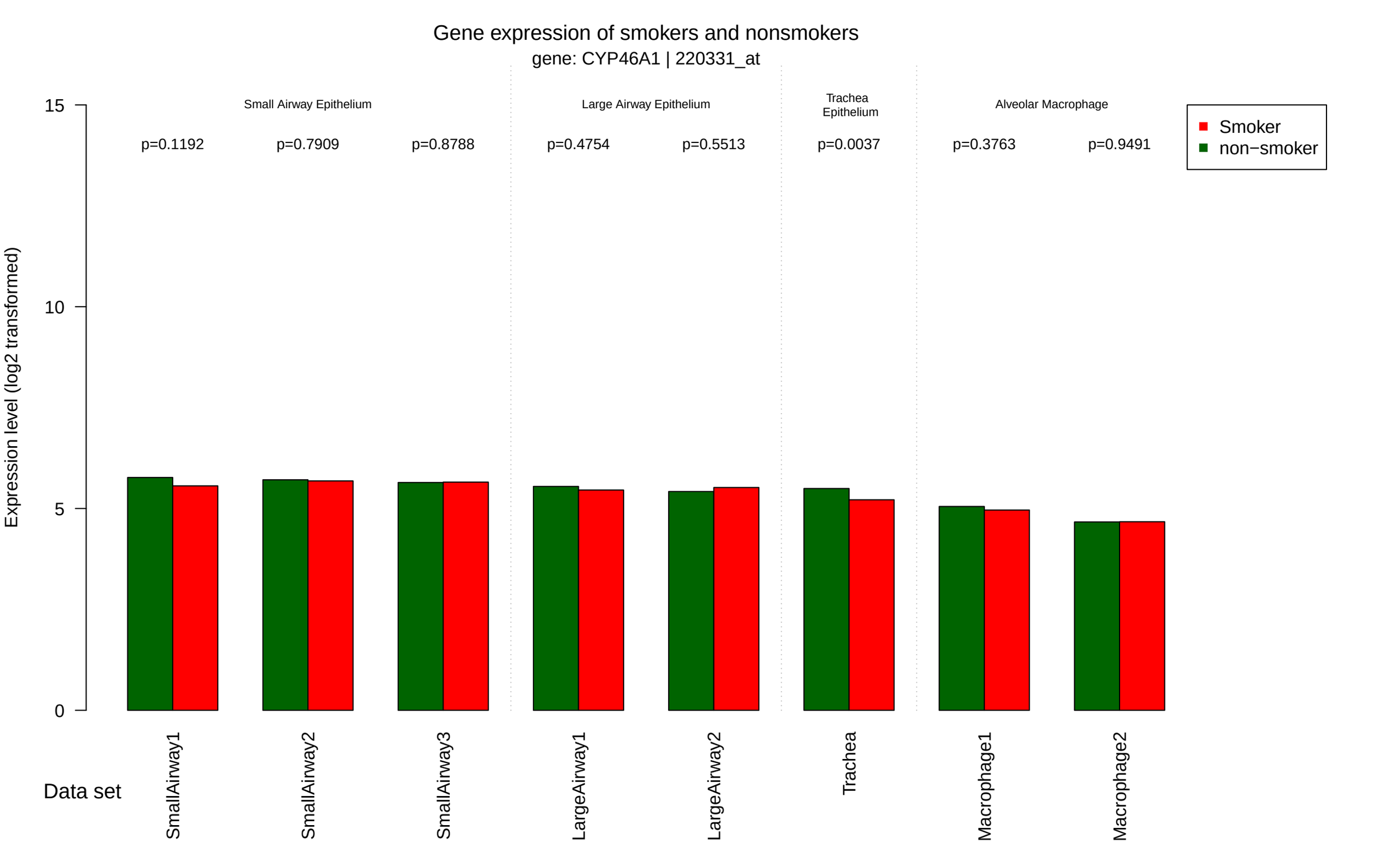
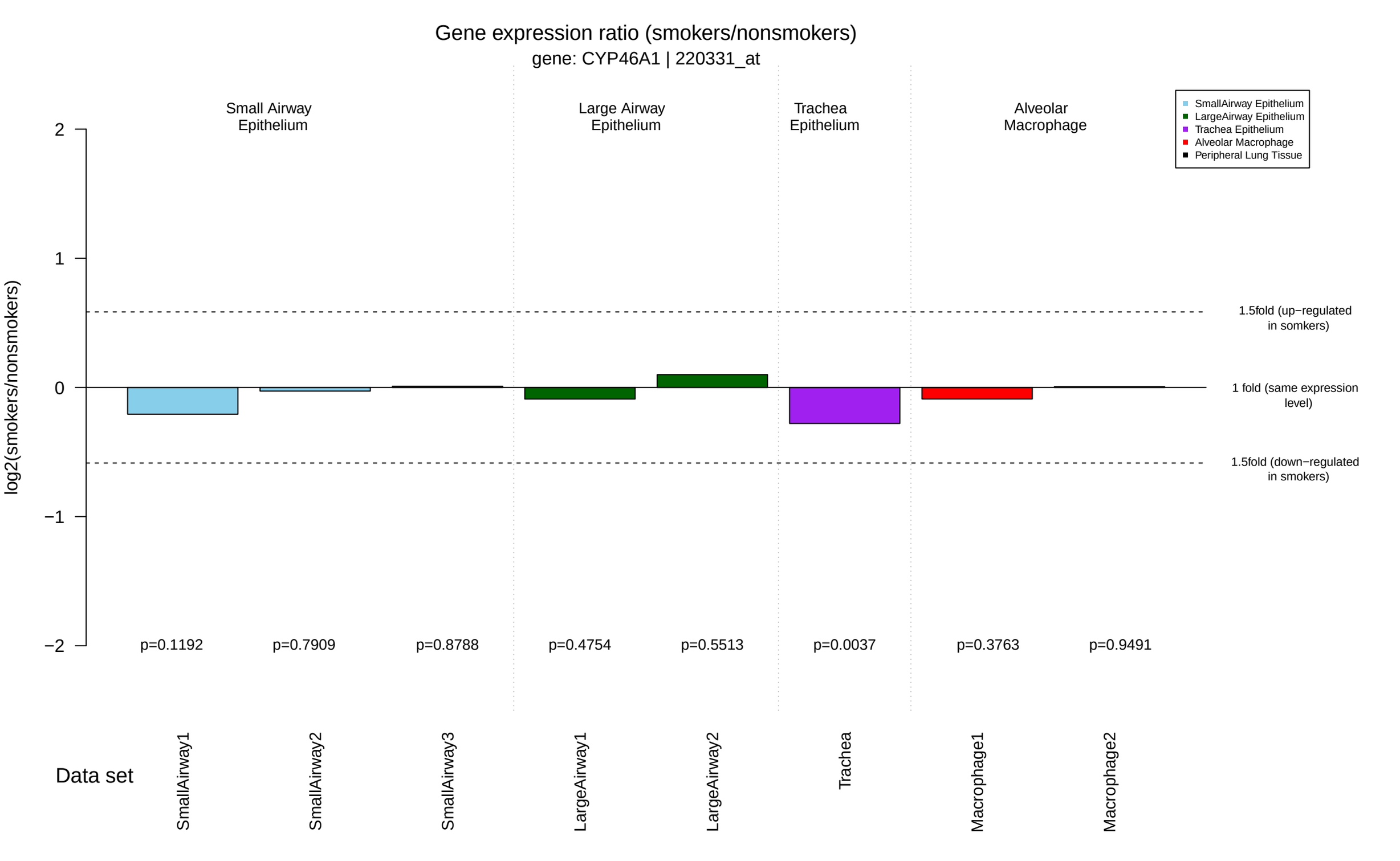
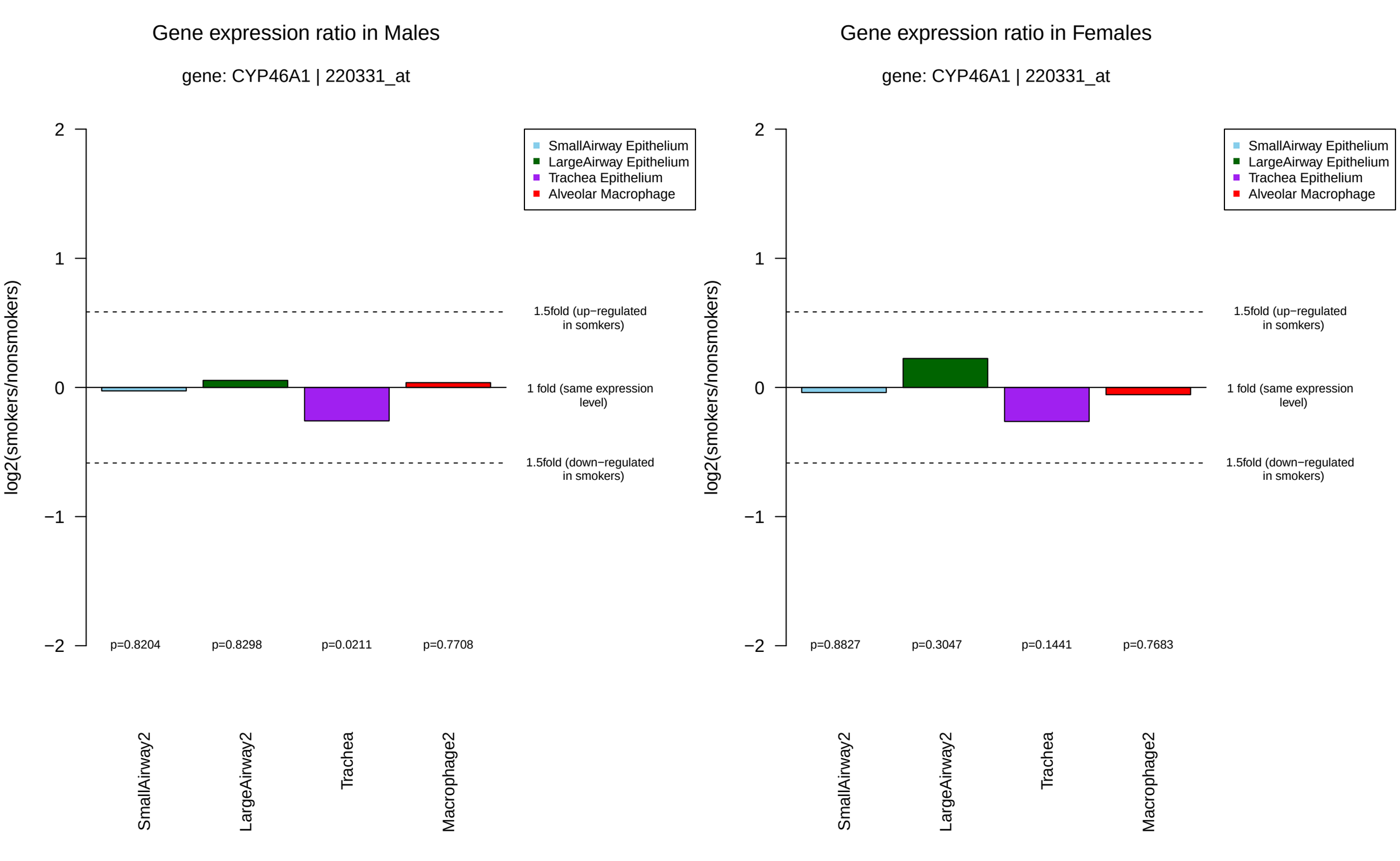
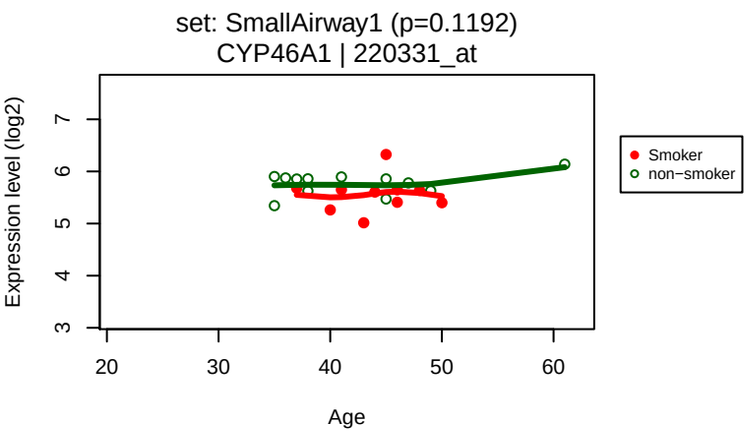
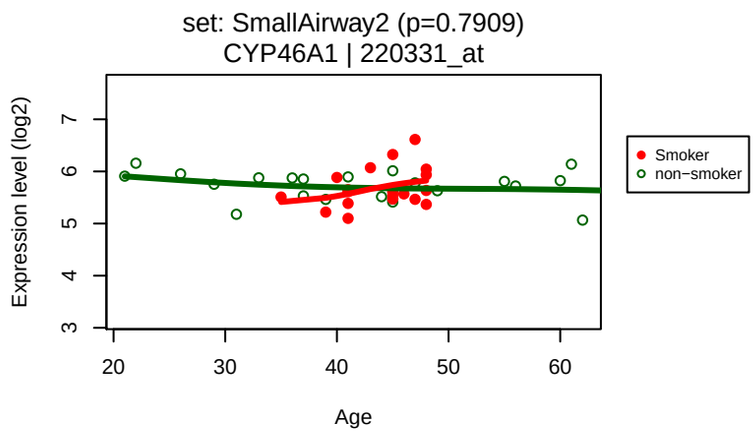
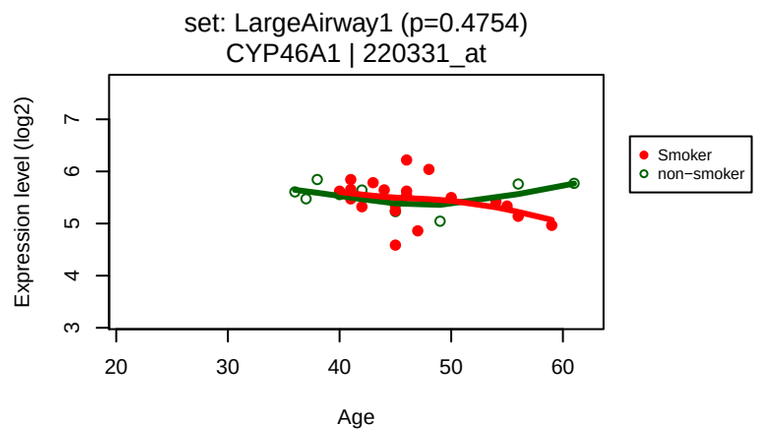

Gene expression of smokers and nonsmokers
gene: CYP46A1 | 220331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1192
p=0.7909
p=0.8788
p=0.4754
p=0.5513
p=0.0037
p=0.3763
p=0.9491
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

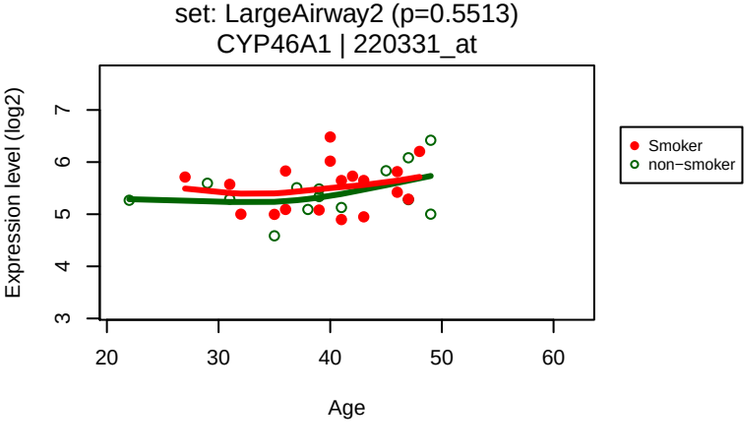

Gene expression ratio (smokers/nonsmokers)
gene: CYP46A1 | 220331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1192
p=0.7909
p=0.8788
p=0.4754
p=0.5513
p=0.0037
p=0.3763
p=0.9491
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

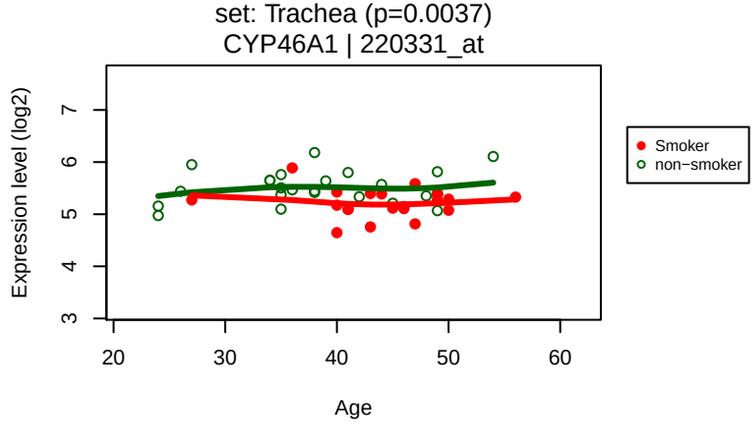

Gene expression ratio in Males
gene: CYP46A1 | 220331_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8204
p=0.8298
p=0.0211
p=0.7708
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP46A1 | 220331_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8827
p=0.3047
p=0.1441
p=0.7683
SmallAirway2
LargeAirway2
Trachea
Macrophage2

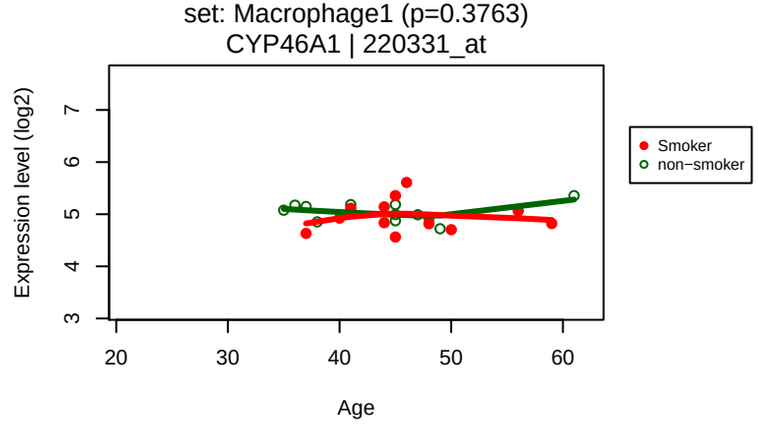

set: SmallAirway1 (p=0.1192)
CYP46A1 | 220331_at
Expression level (log2)
Age
Smoker
non−smoker

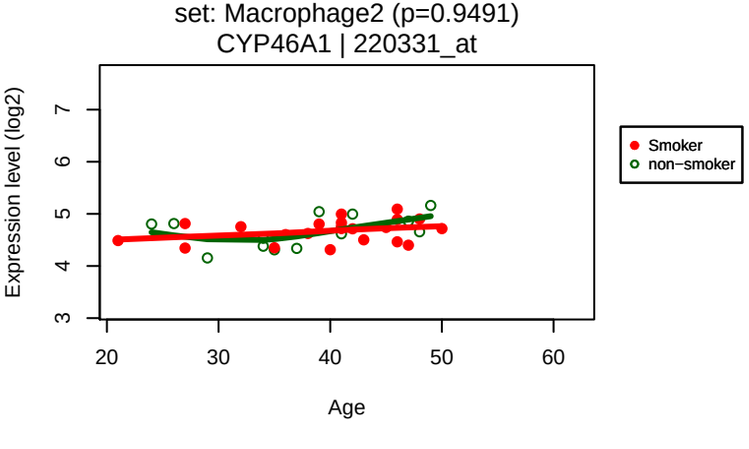

set: SmallAirway2 (p=0.7909)
CYP46A1 | 220331_at
Expression level (log2)
Age
Smoker
non−smoker

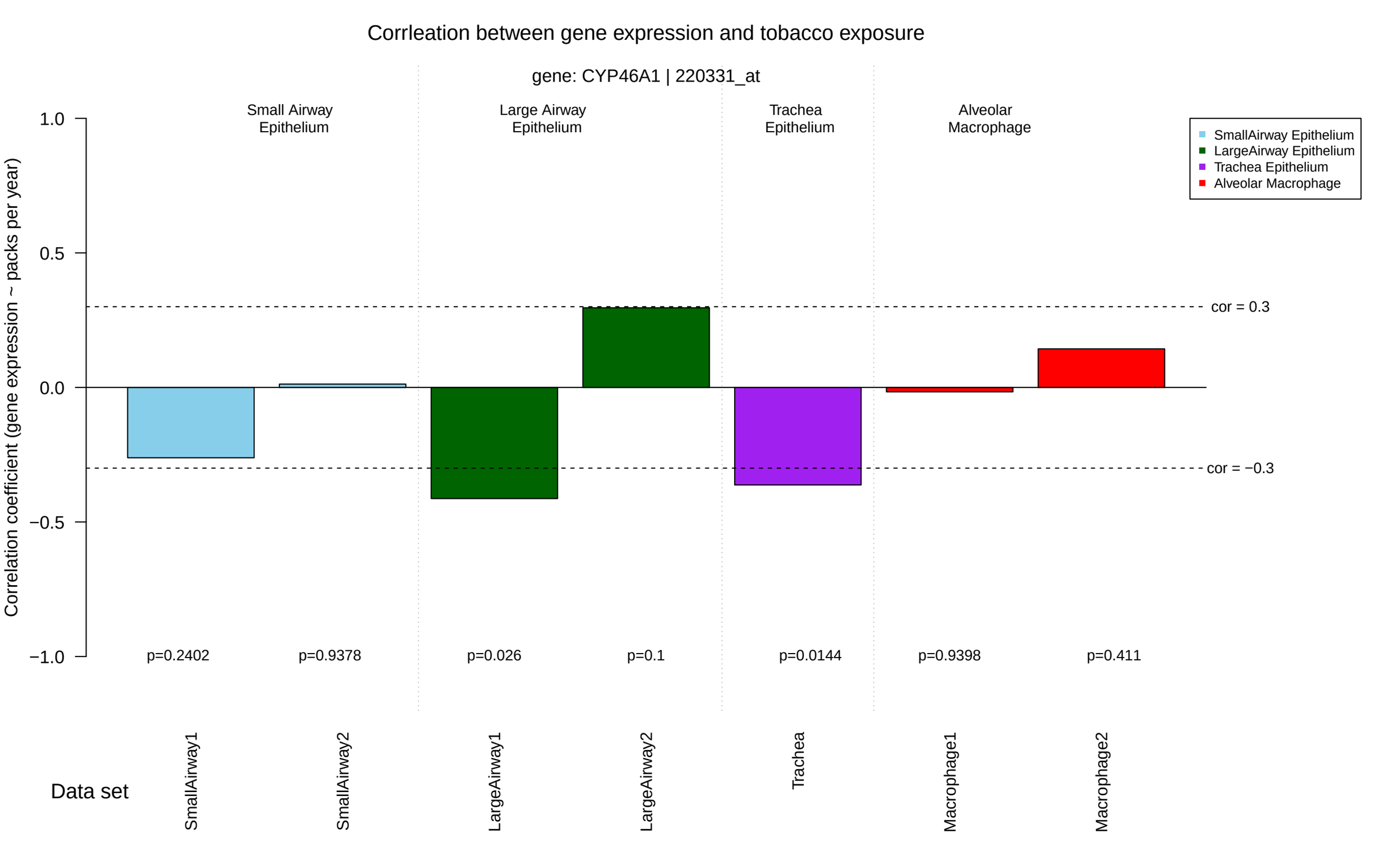

set: LargeAirway1 (p=0.4754)
CYP46A1 | 220331_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5513)
CYP46A1 | 220331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0037)
CYP46A1 | 220331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3763)
CYP46A1 | 220331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9491)
CYP46A1 | 220331_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP46A1 | 220331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2402
p=0.9378
p=0.026
p=0.1
p=0.0144
p=0.9398
p=0.411
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2